nature portfolio

Peer Review File

AMR-GNN: A multi-representation graph neural network framework to enable genomic antimicrobial resistance prediction

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Summary:
The authors have developed a novel AMR phenotype detection tool called AMR-GNN. AMR-GNN utilises a graphical neural network to predict AMR phenotype from several genomic inputs: unitigs, SNPs, and frequency chaos game representation. The authors originally used 2,515 Pseudomonas aeruginosa genomes with corresponding MIC values to 12 antipseudomonal antimicrobials (from broth dilution or agar dilution) to test/train the models. The isolates are from varied sequence types to have a diverse population. They used EUCAST breakpoints to classify their isolates into a binary susceptible or non-susceptible. They trained the models with 80% of the data (~1962 isolates), tested with 20% of the data (~490 isolates), and then used an external validation set of 63 isolates to further test the model predictions. The baseline model using only unitigs is not as good as the complete AMR-GNN model. For example, there is a >28% increase for cefepime when using AMR-GNN. The authors benchmarked their tool against three other AMR prediction methods, and I am not familiar with these methods (see overall comments). I liked that the authors made additional models of key species from both Gram-positive and Gram-negative bacteria. These models also had high predictive accuracy, showing their method is not limited to one species. The authors were able to identify some of the key genes which were making large contributions to the model performance. Many of the genes contributing to the model performance were already known in the AMR databases. They do mention this briefly, but in some sections, this is not clear. They could use this to show that their models are drawing out the correct key genes of interest. However, the authors noted some genes which are not well known and highlighted clearly how they could be involved in contributing to resistance. Overall, I found the paper to be clear and well-written. I found it easy to follow, and the logic between the sections was clear to me. The figures were clear and easy to interpret overall. I wonder if Fig.2b would be easier to visualise in a way that the models for each drug would be easier to compare, for example, a bar chart for each measurement (e.g. F1). However, this could be personal preference. The conclusions lack some thoughts about the significance of the software. For example, identifying possible key genes involved in AMR that are unknown is critical to understanding AMR.

Overall comments:
- The authors often say, “AMR prediction”. I think it would be clearer to say AMR phenotype prediction (so it is not confused with genotype)
- Line 59: The authors mention how WGS approaches will “streamline” the process of AMR detection, but organisms would still have to be cultured to be able to do WGS?
- Lines 117- 118: broth microdilution and agar dilution are very different approaches? The same isolate could have different AMR phenotypes using these two different models. How do you think this will impact the model?
- Line 123-126: I don’t really understand why you selected the external validation data from two small datasets rather than pooling the data and randomly selecting the data. This could have avoided some of the phenotype imbalances.
- Some of the datasets are quite imbalanced, which could cause issues with overfitting.
- Line 146: Would it be possible to give a little bit more detail of what a FCGR is?
- Lines 308-309: Do you think this drop in prediction accuracy could be due to the models being overfit, or that they did not have all the data required to train a model for better predictions?
- I find the benchmarks a bit limited. I am not familiar with any of the benchmark methods; therefore, I think the phrase “state-of-the-art rule-based” is exaggerated, possibly? The authors do not mention tools such as ResFinder, which is much more

highly cited. Do the others not mention these types of tools because they are less complex (i.e. not a neural network model?) or for a different reason?
- Many of the tools do not have version numbers, which are necessary for reproducibility.
- The GitHub for AMR-GNN lacks information about what the model uses to predict and what outputs are expected. I ran the demo data, and the code worked to my knowledge (the Conda command did not work for me and required a bit of tweaking before I could install the software). I have previously built different types of neural networks, so I am familiar with the terminology. However, the output files do not have enough information for someone not familiar with neural networks. I would suggest that the authors provide a summary output file and explain this in the README.md file. The GitHub should explain in detail what kind of input is expected and how the user can prepare their data to be in the correct format. A help menu, either on GitHub or on the command line, would be helpful.
- The paper also explained how the models are biologically interpretable. Could this be explained in the GitHub on how I can interpret the results this way, or is this not currently added into this version?
- Line 441: The authors mention a batch size of 32 for the building of their model. This may be a bit small when training with >2000 genomes. I would like the authors to explain why this is an appropriate method, in case I have misunderstood something.

Line-by-line issues:
- Line 51: “detection of AMR” could be made clearer by saying “detection of AMR phenotype”
- Line 54: “Our current methods for identifying” why not say “Culture-based methods…” to avoid confusion of when people use WGS techniques?
- Line 102: “multiple genome representations” possibly is not the best way to phrase it?
- Line 121: I assume that there are not 12 MIC values for each isolate; it would be nice to say how many MIC values there are in total
- Line 122: “non-sensitive” is used to describe resistant isolates; why not just say “resistant” to avoid confusion about “intermediate” phenotypes being included with non-sensitive?
- Line 148: Why not mention the feature selection here?
- Line 152: Define the acronym “AUROC”
- Lines 180-183: The sentence starting with “Bacteria” is quite a long sentence.
- Line 225: The numbers relating to model performance are confusing.
- Line 239: ‘PATRIC’ is now called BV-BRC
- Lines: 368-369: The authors talk about applying their models to “multi-omics data”, would this be possible if they are taxonomically specific? Or will the models eventually be more flexible this way?

(Remarks on code availability)
The GitHub for AMR-GNN lacks information about what the model uses to predict and what outputs are expected. I ran the demo data, and the code worked to my knowledge (the Conda command did not work for me and required a bit of tweaking before I could install the software). I have previously built different types of neural networks, so I am familiar with the terminology. However, the output files do not have enough information for someone not familiar with neural networks. I would suggest that the authors provide a summary output file and explain this in the README.md file. The GitHub should explain in detail what kind of input is expected and how the user can prepare their data to be in the correct format. A help menu, either on GitHub or on the command line, would be helpful.
The paper also explained how the models are biologically interpretable. Could this be explained in the GitHub on how I can interpret the results this way, or is this not currently added into this version?
Line 441: The authors mention a batch size of 32 for the building of their model. This may be a bit small when training with >2000 genomes. I would like the authors to explain why this is an appropriate method, in case I have misunderstood something.

Reviewer #2

(Remarks to the Author)
The authors introduce AMR-GNN, a dual-graph GNN that uses unitigs as node features and builds two similarity graphs (from SNP and FCGR representations) and fuses them via low-rank multimodal fusion. A simple but effective “decoupling” step drops edges between isolates of the same MLST to blunt clonal-structure bias.
The manuscript is well written, the figures are of top quality, and the supplementary files contain all required information (MIC values and sequence accession IDs). I particularly appreciated the stepwise testing of the algorithm in this study (single features vs full algorithm, with and without decoupling), which makes it clear how each component contributes to overall performance.
Overall, this is an excellent and solid paper that will be of interest to the readership of Nature Communications. I recommend its acceptance for publication if the authors can address the following concerns:

Major concerns

_ benchmarks on other bacterial pathogens: it is impossible to assess the robustness of AMR-GNN performance without reference points (i.e., results obtained from other tools). I would kindly ask the authors to run VAMPr or ARIBA, or any AMR phenotype prediction tool, on these four datasets.

_ content of the discussion section: while I found the overall manuscript very interesting, I’m afraid the Discussion section was particularly poor. The first five paragraphs consist of a long, and perhaps unnecessary, recapitulation of the results. The two paragraphs that follow mention limitations of the study (see below), but there is no real discussion of the analyses. I would have liked some discussion of, for instance, the contribution of each feature type to performance, given their different nature, and which additional ones could be added in the future. Also, how do the authors know that ST is the optimal distance level to remove population effects?

_ applicability of the algorithm: It isn't clear to me after reading the manuscript whether AMR-GNN represents a preliminary attempt at integrating different sources of features or a ready-to-use tool that could be deployed for large-scale AMR inferences. The authors mention two limitations regarding its applicability: the need to rebuild the "database" (I'm afraid I did not understand this part) and its "computational complexity" (are the authors referring to runtime and memory usage?). However, I believe these aspects should be mentioned in the abstract and introduction to let the reader fully understand the current applicability of AMR-GNN.

Minor concerns

_ the GitHub repo could benefit from more detailed instructions regarding AMR-GNN usage. There is currently no manual.
_ supplementary file 1 is a PDF consisting of two tables spanning 270 pages. Could the authors provide these data as an Excel file, which would be easier to read and extract data from?
_ Introduction, page 4: the "high-dimensional nature" of genomic data is not sufficiently explained in the introduction. I personally thought the authors were referring to higher-order features (e.g., impact of variants on promoter activity or protein stability).
_ l. 100–101: "their use in bacterial genomics for AMR prediction remains limited." If GNNs have already been used for AMR prediction, references are needed. If not, the sentence might need slight rewording.
_ l. 286–287: the model does not make any predictions per se of horizontally transferred AMR genes. Please reword.
_ l. 377: please define the abbreviation "BMD."
_ l. 377: please also indicate here how many of these isolates were collected in this study (I believe the number is 341).
_ l. 390: eight, not seven.
_ l. 400: please clarify what "internal" means. If it corresponds to the unpublished set of isolates, then "remaining data" in l. 392 should be modified into "remaining published data."
_ l. 439: consisting "of"?

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)
This manuscript proposes AMR-GNN, a graph deep learning–based framework that integrates multiple genomic representations with graph neural networks (GNN) to predict antimicrobial resistance (AMR) from genomic sequence data of Pseudomonas aeruginosa. Although Figure 1 illustrates that the features are mainly derived from unitigs, SNPs, and FCGR, the description in the Methods section and the demo dataset provided at https://github.com/andyvng/amr-gnn suggest that AMR prediction does not directly start from genomic assemblies, but rather requires an adjacency matrix as input. However, the information on the website appears insufficient to clarify how genomic assemblies can be practically input to obtain AMR predictions. Additional methodological transparency would therefore be beneficial.

Furthermore, while the study employs GNN to evaluate performance using metrics such as F1, AUROC, sensitivity, and specificity, it overlooks what is arguably the most clinically critical metric in AMR prediction: the very major error (VME) rate. In addition, the reported F1 scores (Figure 3) are approximately 0.8, which is lower than those achieved by prior studies (e.g., https://pmc.ncbi.nlm.nih.gov/articles/PMC8575023/), where F1 values around 0.92 were reported (link to model stats). As a result, it is not entirely clear whether this GNN-based approach offers superior performance compared to existing methods.

(Remarks on code availability)
Although I have not reviewed the code at https://github.com/andyvng/amr-gnn, the documentation provided on the website does not include instructions on how to train the AMR-GNN model or how to use genomic sequencing data to perform AMR prediction. Clearer guidance on these steps would greatly enhance the reproducibility and utility of the proposed framework.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
I can see that the authors have clearly listened to myself and the other reviewers to make relevant changes to the manuscript.
The only comments I still have are about the GitHub (see comments on code).

(Remarks on code availability)
comments on code:
I think it would be nice to have a table of each parameter/option used in the command and have a mini explanation of what that option does (even if it seems obvious).

The authors have implemented a Snakemake workflow using a Docker image. However, on some systems, Docker is not feasible (I currently experience this on the HPC I use for work), so you could include instructions on how to build a Singularity container, too. I do not necessarily think this is needed for publication, but it may be nice in the future for people who are less experienced using containers.
I tested the new code, and each stage worked apart from the model interpretation section, this should be fixed before publication. I think it is this line in “explain.py” that is causing the pipeline to fail: baseline = torch.load(cfg.explainer.baseline).to(device).

Reviewer #2

(Remarks to the Author)
The authors have carefully considered my suggestions and revised the manuscript accordingly. The changes have improved the clarity and accuracy of the text. I now recommend the manuscript for publication.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)
As Reviewer #3, my comments primarily focused on the practical usability of the code provided on GitHub. Although the authors state that detailed instructions are available online, I encountered substantial difficulties during installation and was unable to fully test the workflow. In particular, I was not able to use the Docker environment, which prevented me from verifying whether the provided pipeline is functional as described.

Moreover, the current documentation on the GitHub repository (https://github.com/andyvng/amr-gnn
) remains overly simplified and does not sufficiently guide users through installation, input preparation, and execution in a way that ensures reproducibility. Based on my experience, the existing instructions are insufficient to guarantee that an independent user can reliably reproduce or deploy the tool, especially in environments where Docker cannot be used.

(Remarks on code availability)
While the authors indicate that usage instructions are provided on GitHub, I experienced difficulties during installation and was unable to fully evaluate the workflow. As I could not use the Docker environment, I was therefore unable to assess whether the provided code is readily usable in practice.

Dear Reviewers,

Thank you for your careful consideration of our manuscript '*AMR-GNN: A multi-representation graph neural network framework to enable genomic antimicrobial resistance prediction*'. Your comments are much appreciated, and we believe your input has strengthened the manuscript. Please find our point-by-point response to your comments below. All line numbers refer to the manuscript with tracked changes (Simple Markup mode).

REVIEWER #1 (Remarks to the Author)

Comment 1.1:

- The authors often say, "AMR prediction". I think it would be clearer to say AMR phenotype prediction (so it is not confused with genotype)

Response 1.1:

We have replaced "AMR prediction" with "AMR phenotype prediction" as suggested.

Comment 1.2:

- Line 59: The authors mention how WGS approaches will "streamline" the process of AMR detection, but organisms would still have to be cultured to be able to do WGS?

Response 1.2:

We have revised the manuscript accordingly.

Line 61 reads:

"Coupled with advances in bioinformatics, WGS data can now support a range of analyses, including organism identification and AMR detection."

Comment 1.3:

- Lines 117- 118: broth microdilution and agar dilution are very different approaches? The same isolate could have different AMR phenotypes using these two different models. How do you think this will impact the model?

Response 1.3:

Generally, larger training datasets correlate with improved model performance. Although EUCAST prioritises broth microdilution for antimicrobial susceptibility testing (AST), CLSI guidelines suggest that both agar dilution and broth microdilution are suitable for AST in *P. aeruginosa* (except colistin, which was not assessed in our study) [1]. As a result, we incorporated agar dilution data to augment the training set. Notably, these data covered only four antimicrobials (i.e., tobramycin, ceftazidime, ciprofloxacin, and meropenem), achieving strong AUROCs ranging from 0.869 to 0.977. These results suggest that expanding the dataset via the inclusion of agar dilution data enhanced, rather than compromised, model performance.

References

1. CLSI. M100 - Performance standards for antimicrobial susceptibility testing. Clinical and laboratory standards institute Wayne, PA; 2020.

Comment 1.4:

- Line 123-126: I don't really understand why you selected the external validation data from two small datasets rather than pooling the data and randomly selecting the data. This could have avoided some of the phenotype imbalances.

Response 1.14:

The aforementioned steps (i.e., data pooling and random test set selection) were already performed for the internal test set. In this section, our goal was to evaluate how well AMR-GNN generalises to unseen data, while maximising the training data available. Therefore, we selected the two smallest datasets for external validation. We added this clarification to the manuscript.

Line 131 reads:

"We used seven datasets (our centre and [11, 17, 19-21, 23, 24]) for training and internal validation. The two smallest datasets [18, 22] were reserved as a hold-out set to assess the model's generalisability (see Methods)."

Comment 1.5:

- Some of the datasets are quite imbalanced, which could cause issues with overfitting.

Response 1.5:

We thank the reviewer for highlighting the issue of class imbalance. In this study, we calculated class-specific weights and applied them to the binary cross-entropy loss function to account for imbalance. We have updated this in the manuscript.

Line 474 reads:

"To compute the loss, we used binary cross-entropy function (BCELoss) adjusted by class-specific weights."

Comment 1.6:

- Line 146: Would it be possible to give a little bit more detail of what a FCGR is?

Response 1.6:

We have updated the manuscript to provide a more detailed explanation of the FCGR encoding process

Line 154 reads:

"FCGR generates a $2^k \times 2^k$ grid where k-mers are mapped according to their nucleotide sequence and weighted by their occurrence frequency [29,30]. In this representation, each grid cell is analogous to a pixel in an image.

Comment 1.7:

- Lines 308-309: Do you think this drop in prediction accuracy could be due to the models being overfit, or that they did not have all the data required to train a model for better predictions?

Response 1.7:

We agree with the reviewer that the model may lack sufficient data to optimise predictive performance. Given that AMR mechanisms can be geographically specific [1], key resistance determinants present in one region may be absent in another. This likely accounts for why ARDaP surpassed AMR-GNN in predicting resistance to meropenem, ciprofloxacin, and piperacillin/tazobactam in the hold-out test set, as ARDaP may incorporate a broader, globally curated list of AMR mechanisms. We have updated the discussion section to reflect this point.

References:

1. Kocsis B, Gulyás D, Szabó D. Diversity and Distribution of Resistance Markers in *Pseudomonas aeruginosa* International High-Risk Clones. Microorganisms. 2021;9(2):359.

Line 330 reads:

“The observed performance drop in the external test set indicates that larger training datasets are likely required to improve generalisation, particularly for species with complex AMR mechanisms like *P. aeruginosa*. Resistance mechanisms display marked regional specificity, meaning that variants widespread in one location may be absent elsewhere [47]. This likely explains why ARDaP, which uses a comprehensive, globally curated AMR determinant database, outperformed AMR-GNN in 3 out of 6 external comparisons. These results therefore highlight the necessity of continuous data integration and model updating to ensure robustness across different geographical regions and time periods.”

Comment 1.8:

- I find the benchmarks a bit limited. I am not familiar with any of the benchmark methods; therefore, I think the phrase “state-of-the-art rule-based” is exaggerated, possibly? The authors do not mention tools such as ResFinder, which is much more highly cited. Do the others not mention these types of tools because they are less complex (i.e. not a neural network model?) or for a different reason?

Response 1.8:

Per the reviewer’s suggestion, we have been more conservative and removed the term 'state-of-the-art'. The benchmarking methods discussed were specifically designed for AMR phenotype prediction in *P. aeruginosa*. ARDaP, in particular, can be viewed as an enhanced version of ResFinder, integrating both ResFinder’s AMR determinant database and novel *P. aeruginosa*-specific resistance variants identified through GWAS and machine learning analyses. This method has demonstrated superior performance compared to several established AMR phenotype prediction tools, including ResFinder and AMRFinderPlus. We therefore revised the manuscript to clarify the choice of these benchmarks.

Line 218 reads:

“AMR-GNN achieved superior performance compared with *P. aeruginosa*-specific AMR phenotype prediction approaches.”

Line 222 reads:

“We evaluated AMR-GNN against two specific implementations: the method by Cortes-Lara et al. [20], (hereafter ’rule-based’), which utilises the presence/absence of known AMR determinants; and ARDaP [32], an automated scoring system combining known determinants with variants identified via genome-wide association studies (GWAS) and ML-based feature selection. Notably, ARDaP previously outperformed other common AMR phenotype prediction tools (e.g., abritAMR, AMRFinderPlus, and Resfinder) in all 10 tested antimicrobials.”

Comment 1.9:

- Many of the tools do not have version numbers, which are necessary for reproducibility.

Response 1.9:

We have updated the version number for the tools as suggested.

Comment 1.10:

- The GitHub for AMR-GNN lacks information about what the model uses to predict and what outputs are expected. I ran the demo data, and the code worked to my knowledge (the Conda command did not work for me and required a bit of tweaking before I could install the software). I have previously built different types of neural networks, so I am familiar with the terminology. However, the output files do not have enough information for someone not familiar with neural networks. I would suggest that the authors provide a summary output file and explain this in the README.md file. The GitHub should explain in detail what kind of input is expected and how the user can prepare their data to be in the correct format. A help menu, either on GitHub or on the command line, would be helpful.

Response 1.10:

We thank the reviewer for the suggestions. We have implemented Docker containerisation to streamline the environment setup process and improve code reproducibility. We have also developed a Snakemake workflow to generate input data for AMR-GNN from genome assemblies. Furthermore, we have updated the output format to include predicted binary AMR phenotypes, making the results file easier to interpret. Detailed instructions have been added to the GitHub repository. Please see https://github.com/andyvng/amr-gnn.

Comment 1.11:

- The paper also explained how the models are biologically interpretable. Could this be explained in the GitHub on how I can interpret the results this way, or is this not currently added into this version?

Response 1.11:

We have updated the GitHub repository to include instructions on running model interpretation, as suggested. Please refer to the **Model interpretation** section.

Comment 1.12:

- Line 441: The authors mention a batch size of 32 for the building of their model. This may be a bit small when training with >2000 genomes. I would like the authors to explain why this is an appropriate method, in case I have misunderstood something.

Response 1.12:

While the dataset comprised of 2,515 genomes in total, label sparsity remains an issue for some antimicrobials (with as few as 601 entries for cefepime). Therefore we used mini-batch gradient descent to maintain training stability. We selected a batch size of 32, which is established in the literature as an effective default. We have updated the manuscript with references supporting this hyperparameter choice.

Line 475 reads:

"We employed mini-batch gradient descent with a batch size of 32, which is regarded as a robust empirical baseline [60, 61]."

Comment 1.13:

- Line 51: “detection of AMR” could be made clearer by saying “detection of AMR phenotype”

Response 1.13:

We have updated the manuscript accordingly.

Line 51 reads:

“Rapid and accurate detection of AMR phenotype through antimicrobial susceptibility testing (AST) is crucial for improving patient outcomes”

Comment 1.14:

- Line 54: “Our current methods for identifying” Why not say “Culture-based methods…” to avoid confusion of when people use WGS techniques?

Response 1.14:

We have updated the manuscript as suggested.

Line 54 reads:

“Culture-based methods for identifying AMR are slow, laborious and are often not reported to the clinician for 2-5 days after sample collection”

Comment 1.15:

- Line 102: “multiple genome representations” possibly is not the best way to phrase it?

Response 1.15:

We have replaced “multiple genome representations” with “multiple WGS-derived features”.

Line 102 reads:

“We therefore developed AMR-GNN, a new GNN-based framework using multiple WGS-derived features to predict AMR phenotype in *P. aeruginosa.*”

Comment 1.16:

- Line 121: I assume that there are not 12 MIC values for each isolate; it would be nice to say how many MIC values there are in total

Response 1.16:

We have added the total number of MIC values as suggested.

Line 126 reads:

“The AST dataset consisted of 19,865 MIC entries covering 12 common antipseudomonal antimicrobials.”

Comment 1.17:

- Line 122: “non-sensitive” is used to describe resistant isolates; why not just say “resistant” to avoid confusion about “intermediate” phenotypes being included with non-sensitive?

Response 1.17:

We have replaced “non-sensitive” with “resistant” as suggested.

Comment 1.18:

- Line 148: Why not mention the feature selection here?

Response 1.18:

In this initial analysis, the entire feature sets were provided to the elastic net models, allowing the algorithm to identify the most predictive features. We have updated the manuscript accordingly.

Line 158 reads:

“The complete feature sets for unitigs and SNPs contained approximately 1.5 million and 900,000 features, respectively, which were used as input into the elastic net.”

Comment 1.19:

- Line 152: Define the acronym “AUROC”

Response 1.19:

We have defined AUROC as suggested.

Line 164 reads:
“area under receiver operating characteristic curve (AUROC)”

Comment 1.20:
- Lines 180-183: The sentence starting with “Bacteria” is quite a long sentence.

Response 1.20:
We have rewritten the sentence as suggested.

Line 192 reads:
“Bacterial population structure is a major confounder in AMR prediction because of slow linkage disequilibrium decay [31]. This results in co-inheritance of causal and non-causal variants that ML models cannot easily disentangle.”

Comment 1.21:
- Line 225: The numbers relating to model performance are confusing.

Responses 1.21:
We have rewritten the sentences accordingly.

Line 235 reads:
“Based on label availability, we benchmarked AMR-GNN against a rule-based approach, ARDaP, and VAMPr using both internal and external test sets (**Fig. 3**). On the internal test set, AMR-GNN achieved universally higher F1-scores and better very major error rates (VME) across all tested antimicrobials (n=5, 8, and 8, respectively). In the external validation (comprising 5, 6, and 6 antimicrobials), AMR-GNN demonstrated superior F1-scores in 4/5, 3/6, and 4/6 cases, respectively. For VME, AMR-GNN achieved better results for 5/5, 4/6, and 4/6 cases, respectively.”

Comment 1.22:
- Line 239: ‘PATRIC’ is now called BV-BRC

Response 1.22:

We thank the reviewer for pointing this out. We have replaced PATRIC with BV-BRC accordingly.

Line 252 reads:

“The analysis covered a comprehensive panel of antimicrobials specific to each species, sourced from The Bacterial and Viral Bioinformatics Resource Center (BV-BRC) databases [37, 38]”

Comment 1.23:

- Lines: 368-369: The authors talk about applying their models to “multi-omics data”, would this be possible if they are taxonomically specific? Or will the models eventually be more flexible this way?

Response 1.23:

Although the current implementation uses WGS-derived representations to construct node and edge features, the framework is inherently flexible and allows for the substitution of these inputs with other ‘omics data. We have cited studies demonstrating the efficacy of non-genomic data in AMR phenotype prediction, suggesting that their integration could further enhance predictive power. However, the scarcity of datasets encompassing all such modalities simultaneously remains a limiting factor, presenting a significant opportunity for future exploration.

Line 399 reads:

“AMR-GNN presents an interpretable and flexible framework for such integration, with promising potential for application to multi-omics modalities such as proteomics and transcriptomics, which have also demonstrated utility in predicting AMR phenotype [11, 51, 52].”

REVIEWER #1 (Remarks on code availability)

Comment 1.24:

The GitHub for AMR-GNN lacks information about what the model uses to predict and what outputs are expected. I ran the demo data, and the code worked to my knowledge (the Conda command did not work for me and required a bit of tweaking before I could install the software). I have previously built different types of neural networks, so I am familiar with the terminology. However, the output files do not have enough information for someone not familiar with neural networks. I would suggest that the authors provide a summary output file and explain this in the README.md file. The GitHub should explain in detail what kind of input is expected and how the user can prepare their data to be in the correct format. A help menu, either on GitHub or on the command line, would be helpful.

Response 1.24:

This comment overlaps with comment 1.10. Please refer to response 1.10.

Comment 1.25:

The paper also explained how the models are biologically interpretable. Could this be explained in the GitHub on how I can interpret the results this way, or is this not currently added into this version?

Response 1.25:

This comment overlaps with comment 1.11. Please refer to response 1.11.

Comment 1.26:

Line 441: The authors mention a batch size of 32 for the building of their model. This may be a bit small when training with >2000 genomes. I would like the authors to explain why this is an appropriate method, in case I have misunderstood something.

Response 1.26:

This comment overlaps with comment 1.12. Please refer to response 1.12.

REVIEWER #2 (Remarks to the Author)

Comment 2.1:

_ benchmarks on other bacterial pathogens: it is impossible to assess the robustness of AMR-GNN performance without reference points (i.e., results obtained from other tools). I would kindly ask the authors to run VAMPr or ARIBA, or any AMR phenotype prediction tool, on these four datasets.

Response 2.1:

We thank the reviewer for the suggestion. We have now expanded our analysis to include a comparison between AMR-GNN and ResFinder across all other bacterial pathogens. ResFinder was selected as the benchmark due to its extensive antimicrobial coverage and superior performance on divergent genomic datasets, where it frequently surpasses other approaches, including ML-based tools [1]. This makes it particularly suitable for benchmarking against the globally sourced BV-BRC dataset. Overall, AMR-GNN outperformed ResFinder in the majority of species–antimicrobial combinations. We have updated the manuscript accordingly.

References:

1. Hu K, Meyer F, Deng Z-L, Asgari E, Kuo T-H, Münch PC, et al. Assessing computational predictions of antimicrobial resistance phenotypes from microbial genomes. Briefings in Bioinformatics. 2024;25(3). doi: 10.1093/bib/bbae206.

Line 264 reads:

"In terms of benchmarking, we compared the performance of AMR-GNN with ResFinder, a tool that has shown strong results relative to several other AMR phenotype prediction methods [5]. Specifically, we assessed predictions across 16 species-antimicrobial combinations. In terms of F1-score, AMR-GNN outperformed ResFinder in 14 of these 16 comparisons. The only exceptions were tobramycin in *E. coli* (0.834 vs. 0.859) and fusidic acid in *S. aureus* (0.929 vs. 0.969) (**Fig. 4i**). The most significant performance improvement with AMR-GNN was observed for linezolid in *E. faecium* (0.859 vs 0.512)."

Comment 2.2:

_ content of the discussion section: while I found the overall manuscript very interesting, I'm afraid the Discussion section was particularly poor. The first five paragraphs consist of a long, and perhaps unnecessary, recapitulation of the results. The two paragraphs that follow mention limitations of the study (see below), but there is no real discussion of the analyses. I would have liked some discussion of, for instance, the contribution of each feature type to performance, given their different nature, and which additional ones could be added in the future. Also, how do the authors know that ST is the optimal distance level to remove population effects?

Response 2.2:

We thank the reviewer for the suggestions to improve the Discussion part. Accordingly, we have condensed the section to improve clarity and flow. Please refer to the revised Discussion section in the manuscript.

Regarding the reviewer's specific suggestions, we first discussed how feature selection might affect model performance. As multi-omics data are increasingly collected, we provide recommendations on how these data can be integrated into the models and what features may warrant inclusion.

Line 318 reads:

"Consistent with previous work [7], we found that reference-free genomic features (i.e., unitigs) achieved the best performance among single-feature models. However, we also observed that incorporating pairwise genomic distances, which are widely used for detecting outbreaks [46], as edge features significantly improved predictive power. This multi-view integration likely accounts for the model's superior performance over single-feature models and *P. aeruginosa*-specific rule-based approaches across all tested antipseudomonal agents. These findings encourage the integration of additional data types to further enhance performance. For instance, although it is costly to obtain, incorporating transcriptomic data as node features warrants consideration for *P. aeruginosa* given its capacity for environment-driven transcriptional adaptation [47]."

Finally, we acknowledge the limitation of the use of MLST to account for population structure.

Line 383 reads:

"Third, while our approach yielded performance improvements, we acknowledge the inherent limitations of using ST to account for population structure bias. While ST data is widely available and generally informative, its dependence on a limited set of housekeeping genes disregards significant genomic variation., Future iterations could therefore benefit from employing higher-resolution typing schemes, such as core-genome MLST (cgMLST), to better address lineage biases.”

Comment 2.3:

_ applicability of the algorithm: It isn’t clear to me after reading the manuscript whether AMR-GNN represents a preliminary attempt at integrating different sources of features or a ready-to-use tool that could be deployed for large-scale AMR inferences. The authors mention two limitations regarding its applicability: the need to rebuild the “database” (I’m afraid I did not understand this part) and its “computational complexity” (are the authors referring to runtime and memory usage?). However, I believe these aspects should be mentioned in the abstract and introduction to let the reader fully understand the current applicability of AMR-GNN.

Response 2.3:

We thank the reviewer for the suggestion. We have revised the abstract and introduction to explicitly frame this study as a proof-of-concept. Our objective is to demonstrate the potential of a graph neural network using multi-representation inputs to enhance AMR phenotype prediction relative to single-feature approaches. Although the current iteration relies exclusively on WGS-derived features, the framework is designed for multimodal learning and is capable of integrating other ‘omics layers (e.g., proteomic or transcriptomic) when those data become available.

Line 40 reads:

“We present AMR-GNN as a proof-of-concept framework designed to address several key problems in AMR phenotype prediction with machine learning (ML) approaches, including using multiple genomic representations to enhance performance, to mitigate the influence of clonal relationships and to identify informative biomarkers to provide explainability.”

Line 113 reads:

“Collectively, these results demonstrate the potential of leveraging distinct feature types to enhance AMR phenotype prediction, establishing AMR-GNN as a robust proof-of-concept framework for integrating these features into a unified ML architecture.”

We have revised the manuscript to clarify the need to reconstruct the unitig database.

Line 376 reads:
“While new isolates can be queried against an existing unitig list, this approach may obscure novel biological signals. For instance, if two isolates have distinct mutations at the same nucleotide position, the reference unitig would be recorded as “absent” for both. Without updating the database to include the specific unitigs for these new variants, the genetic nuance distinguishing these isolates is lost.”

Finally, the term “computational complexity'” refers to the preprocessing step required to generate feature inputs for AMR-GNN, specifically the computation of pairwise distance matrices, where the cost increases quadratically with each added isolate. However, this constraint was primarily relevant to the benchmarking against simpler rule-based approaches for other pathogens (i.e., *E. coli*, *K. pneumoniae*, *S. aureus*, *E. faecium*). As we addressed this in Response 2.1, we have removed this specific discussion from the manuscript to avoid confusion.

Comment 2.4:
_ the GitHub repo could benefit from more detailed instructions regarding AMR-GNN usage. There is currently no manual.

Response 2.4:
We have updated the GitHub repository to include detailed instructions for using AMR-GNN. This documentation covers the pre-processing pipeline for genome assemblies as well as the Docker container for executing the code. Please see https://github.com/andyvng/amr-gnn

Comment 2.5:
_ supplementary file 1 is a PDF consisting of two tables spanning 270 pages. Could the authors provide these data as an Excel file, which would be easier to read and extract data from?

Response 2. 5:

Sorry for the inconvenience. Supplementary File 1 was originally uploaded as an Excel file but was automatically converted to PDF by the submission system. We have provided a download link for the original Excel file below.

https://docs.google.com/spreadsheets/d/1_wXfO-HSKPXO3oajZLqvW79DOtbRompS

Comment 2.6:

_ Introduction, page 4: the "high-dimensional nature" of genomic data is not sufficiently explained in the introduction. I personally thought the authors were referring to higher-order features (e.g., impact of variants on promoter activity or protein stability).

Response 2.6:

In this context, 'high-dimensional' refers to the sequence of millions of nucleotides comprising bacterial WGS data (e.g., approximately 6 Mb for *P. aeruginosa*). We have revised this section of the manuscript to ensure this interpretation is clear.

Line 79 reads:

"Given that bacterial genomes consist of sequences containing millions of nucleotides (e.g., approximately 6 Mb for *P. aeruginosa*), WGS data can be represented in multiple ways for input into ML models."

Comment 2.7:

_ l. 100–101: "their use in bacterial genomics for AMR prediction remains limited." If GNNs have already been used for AMR prediction, references are needed. If not, the sentence might need slight rewording.

Response 2.7:

We have added the reference as suggested.

Line 101 reads:

"their use in bacterial genomics for AMR phenotype prediction remains limited [15]."

Comment 2.8:

_ l. 286–287: the model does not make any predictions per se of horizontally transferred AMR genes. Please reword.

Response 2.8:

We have updated the manuscript accordingly

Line 306 reads:

“This enables our model to not only generate informative AMR embeddings but also capture shared genetic content among bacterial isolates.”

Comment 2.9:

_ l. 377: please define the abbreviation “BMD.”

Response 2.9

We have added definition for BMD

Line 410 reads:

“broth microdilution (BMD)”

Comment 2.10:

_ l. 377: please also indicate here how many of these isolates were collected in this study (I believe the number is 341).

Response 2.10:

We have added the detail as suggested

Line 413 reads:

“For our in-house collection (n=341), BMD was performed using the SensititreTM Gram negative plate (ThermoFisher, plate code: DKMNG).”

Comment 2.11:

_ l. 390: eight, not seven.

Response 2.11:

We have corrected the typo accordingly.

Line 423 reads:

“Consequently, we conducted external validation for eight antimicrobials”

Comment 2.12:

_ l. 400: please clarify what “internal” means. If it corresponds to the unpublished set of isolates, then “remaining data” in l. 392 should be modified into “remaining published data.”

Response 2.12:

We have revised the manuscript to clarify the distinction between the internal and hold-out test sets, with changes marked in red.

Line 418 reads:

“First, for external validation, we identified two collections with the smallest sample sizes (42 and 21 isolates) and set them aside as a hold-out set (referred to as the “external test set”) [18, 22]. The hold-out dataset contained AST results for nine antimicrobials; however, we excluded imipenem from the external validation due to absolute resistance rate (100%). Consequently, we conducted external validation for eight antimicrobials: meropenem, tobramycin, amikacin, ciprofloxacin, ceftolozane/tazobactam, ceftazidime, piperacillin/tazobactam, and aztreonam. We split the remaining data into training and testing sets using an 80/20 stratified split based on AST labels for each antimicrobial. This test set is referred to as the “internal test set”. To account for performance variability, this process was repeated 10 times with random seeds. For consistency, the same training/testing splits were used across all experiments, including models based on single genomic representations, AMR-GNN training, and benchmarking with external AMR phenotype prediction tools. For AMR-GNN and the CNN model using FCGR inputs, the training set was further divided into training (80%) and validation (20%) subsets. These models were trained on the reduced training set to minimise validation loss. The final models were evaluated on both internal and external test sets.”

Comment 2.13:

_ l. 439: consisting “of”?

Response 2.13:

We have corrected the typo accordingly.

Line 473 reads:

“consisting of 3 convolutional layers and a final classification head.”

REVIEWER #3 (Remarks to the Author)

Comment 3.1:

This manuscript proposes AMR-GNN, a graph deep learning–based framework that integrates multiple genomic representations with graph neural networks (GNN) to predict antimicrobial resistance (AMR) from genomic sequence data of Pseudomonas aeruginosa. Although Figure 1 illustrates that the features are mainly derived from unitigs, SNPs, and FCGR, the description in the Methods section and the demo dataset provided at https://github.com/andyvng/amr-gnn suggest that AMR prediction does not directly start from genomic assemblies, but rather requires an adjacency matrix as input. However, the information on the website appears insufficient to clarify how genomic assemblies can be practically input to obtain AMR predictions. Additional methodological transparency would therefore be beneficial.

Response 3.1:

We have implemented a pre-processing workflow to transform genome assemblies into model-compatible feature inputs. Please see https://github.com/andyvng/amr-gnn

Comment 3.2:

Furthermore, while the study employs GNN to evaluate performance using metrics such as F1, AUROC, sensitivity, and specificity, it overlooks what is arguably the most clinically critical metric in AMR prediction: the very major error (VME) rate.

Response 3.2:

We thank the reviewer for pointing out VME metrics. We have now added results for both VME and major error rates (ME). Please see **Supp. Table 3** and **Supp. Table 8**. We have also revised the manuscript accordingly.

Line 486 reads:

In addition, we also reported clinically related metrics, i.e., VME and ME:

$$Very\ major\ error\ (VME) = \frac{Resistant\ isolates\ falsely\ predicted\ as\ susceptible}{Total\ number\ of\ resistant\ isolates}$$

$$Major\ error\ (VME) = \frac{Susceptible\ isolates\ falsely\ predicted\ as\ resistant}{Total\ number\ of\ susceptible\ isolates}$$

The optimal threshold was determined by minimising the weighted sum of VME and ME:

$$Optimal\ threshold\ = argmin(\lambda VME + ME)$$

Given that the Food and Drug Administration (FDA) establishes stricter acceptable thresholds for VME (1.5%) compared to ME (3%) for new AST systems [60], we applied a weighting factor of λ=2 to prioritise VME minimisation.

Line 236 reads:

"On the internal test set, AMR-GNN achieved universally higher F1-scores and better very major error rates (VME) across all tested antimicrobials (n=5, 8, and 8, respectively). In the external validation (comprising 5, 6, and 6 antimicrobials), AMR-GNN demonstrated superior F1-scores in 4/5, 3/6, and 4/6 cases, respectively. For VME, AMR-GNN achieved better results for 5/5, 4/6, and 4/6 cases, respectively."

Line 263 reads:

"VME values ranged from 0.008 to 0.133, while ME values ranged from 0.054 to 0.336."

Line 350 reads:

"Although VME and ME are critical for clinical translation, they are rarely detailed in AMR phenotype prediction literature. Compared to reported values for *K. pneumoniae* and *S. aureus* [48], our model achieved better VME rates in most cases (5/6 and 3/3, respectively). The mixed performance regarding ME (superior in 2/6 and 1/3 comparisons) is attributable to our threshold optimisation process, which assigned a higher penalty weight to VME."

Comment 3.3:

In addition, the reported F1 scores (Figure 3) are approximately 0.8, which is lower than those achieved by prior studies (e.g., https://pmc.ncbi.nlm.nih.gov/articles/PMC8575023/), where F1 values around 0.92 were reported (link to model stats). As a result, it is not entirely clear whether this GNN-based approach offers superior performance compared to existing methods.

Response 3.3:

We thank the reviewer for highlighting the average F1 scores from previous studies. However, a precise head-to-head comparison is difficult, as the cited study presents these values only as a color-coded heatmap (Figure 5) without providing raw numerical data. Based on the colour scale, the high mean F1 scores reported appear to be driven by species not included in our dataset, such as *Campylobacter jejuni*, *Clostridium difficile*, and *Neisseria gonorrhoeae*. Conversely, the performance for *P. aeruginosa* appears to fall within the average range. To address the broader context, we have included F1 scores for additional species from the BV-BRC database in **Supp. Table 8**. These results demonstrate that AMR-GNN achieves strong performance across these additional species, reaching F1 scores as high as 0.979 for *K. pneumoniae*-ceftriaxone (please see **Supp. Table 8**).

REVIEWER #3 (Remarks on code availability):

Comment 3.4:

Although I have not reviewed the code at https://github.com/andyvng/amr-gnn, the documentation provided on the website does not include instructions on how to train the AMR-GNN model or how to use genomic sequencing data to perform AMR prediction. Clearer guidance on these steps would greatly enhance the reproducibility and utility of the proposed framework.

Responses 3.4:

We have updated the Github repository to include instructions to train AMR-GNN and perform prediction. Please see https://github.com/andyvng/amr-gnn.

Dear Reviewers,

Thank you for your careful consideration of our manuscript '*AMR-GNN: A multi-representation graph neural network framework to enable genomic antimicrobial resistance prediction*'. Your comments are much appreciated, and we believe your input has strengthened the manuscript. Please find our point-by-point response to your comments below.

REVIEWER #1:

Comment 1.1:

I think it would be nice to have a table of each parameter/option used in the command and have a mini explanation of what that option does (even if it seems obvious).

Response 1.1:

We have added tables with argument and brief description to instruct how to train the model to the Github repository. Please see https://github.com/andyvng/amr-gnn.

Comment 1.2:

The authors have implemented a Snakemake workflow using a Docker image. However, on some systems, Docker is not feasible (I currently experience this on the HPC I use for work), so you could include instructions on how to build a Singularity container, too. I do not necessarily think this is needed for publication, but it may be nice in the future for people who are less experienced using containers.

Response 1.2:

We thank the reviewer for the helpful suggestion. Because Apptainer (formerly Singularity) images can be built from Docker images, we have now included instructions on how to build the Apptainer image and how to run the training and prediction workflows. Please see https://github.com/andyvng/amr-gnn.

Comment 1.3:

I tested the new code, and each stage worked apart from the model interpretation section, this should be fixed before publication. I think it is this line in "explain.py" that is causing the pipeline to fail: baseline = torch.load(cfg.explainer.baseline).to(device).

Response 1.3:

We thank the reviewer for pointing out this issue. The code has now been corrected accordingly.

REVIEWER #3

Comment 3.1:

As Reviewer #3, my comments primarily focused on the practical usability of the code provided on GitHub. Although the authors state that detailed instructions are available online, I encountered substantial difficulties during installation and was unable to fully test the workflow. In particular, I was not able to use the Docker environment, which prevented me from verifying whether the provided pipeline is functional as described.

Moreover, the current documentation on the GitHub repository (https://github.com/andyvng/amr-gnn
) remains overly simplified and does not sufficiently guide users through installation, input preparation, and execution in a way that ensures reproducibility. Based on my experience, the existing instructions are insufficient to guarantee that an independent user can reliably reproduce or deploy the tool, especially in environments where Docker cannot be used.

While the authors indicate that usage instructions are provided on GitHub, I experienced difficulties during installation and was unable to fully evaluate the workflow. As I could not use the Docker environment, I was therefore unable to assess whether the provided code is readily usable in practice.

Response 3.1:

We thank the reviewer for evaluating our codebase. We have now added instructions for running the analysis locally using the `uv` package. Because we provide a locked environment file, `uv` can automatically create the Python environment on first use and reuse it on the local machine. We hope this resolves the installation issues and enables the analysis to run smoothly. Please see the ‘Run with uv’ sections in our GitHub repository.

Additionally, as requested by Reviewer #1, we have included a configuration table for each command. We hope this further clarifies the instructions and supports reproducibility of the workflow.